Major Project Synopsis

MCA IV Semester
Department of Computer Science, Jamia Millia Islamia
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1. Project Title

SemantiCite.ai

An Agentic System for Semantic Citation Analysis using Graph RAG

2. Introduction

The volume of scientific literature is increasing rapidly, making it difficult for researchers to trace the genealogy of ideas or understand the specific nature of relationships between research papers. Traditional search engines (like Google Scholar or IEEE Xplore) provide list-based results that treat all citations equally. They indicate *that* Paper A cited Paper B, but they fail to explain *why*.

SemantiCite.ai is an advanced AI-driven system designed to solve this problem. Unlike standard tools that simply visualize citation links, this project employs **Agentic AI** and **Graph Retrieval-Augmented Generation (Graph RAG)** to semanticize citation networks. The system autonomously reads the context of citations (e.g., distinguishing whether a paper *supports*, *refutes*, or *extends* prior work) and builds an interactive, reasoning-capable Knowledge Graph.

This project aims to provide a next-generation research assistant that helps scholars visualize the "evolution of truth" in their domain.

3. Problem Statement

Current research tools suffer from a "Semantic Blindness" regarding citations:

a) **Lack of Context:** A citation is treated as a binary link. Researchers cannot easily distinguish between a "foundational" citation and a "critique" without manually reading the full text.
b) **Information Overload:** A keyword search often yields thousands of disconnected results, making it hard to identify the "root" paper or the most "controversial" paper in a specific field.
c) **Static Retrieval:** Existing RAG systems (Retrieval-Augmented Generation) retrieve text chunks based on similarity but fail to understand the structural and relational depth of scientific arguments.

There is a need for a system that can "read" the literature recursively and map not just the papers, but the **nature of the arguments** connecting them

4. Objectives

- **To develop an AI Agent** capable of autonomously searching and retrieving research papers using the Semantic Scholar API.
- **To implement a "Citation Context Extractor"** that identifies the specific sentences where a reference is made within the source text.
- **To build a Semantic Classifier** using Large Language Models (LLMs) that categorizes relationships into distinct types (e.g., *Support, Critique, Method Usage, Basis*).
- **To construct a Knowledge Graph (Neo4j)** that stores papers as nodes and semantic relationships as directed, labeled edges.
- **To create an Interactive Visualization Dashboard** (using React Flow/Cytoscape) that allows users to visually explore the genealogy of research topics.

5. Proposed Methodology / Approach

The project follows the Agile Development Methodology, allowing for iterative refinement of the AI agent's performance.

5.1. System Architecture: The system is built on a modern 3-Tier Architecture

- Frontend (Presentation Layer): A React-based Single Page Application (SPA) for query input and graph visualization.
- Backend (Application Layer): A FastAPI server that hosts the "Agent Orchestrator" (built with LangGraph). This layer handles the logic of fetching papers, parsing text, and calling the LLM.
- Database (Data Layer): A Neo4j Graph Database to store the structured knowledge (Nodes: Papers; Edges: Relationships).

5.2. Data Flow Process

1. Input: User enters a topic (e.g., "Attention Mechanism").
2. Search & Retrieve: The Agent fetches the top seed papers via API.
3. Context Extraction: The system parses the full text to find the "citation window" (the 1-2 sentences surrounding a reference).
4. LLM Reasoning: The Agent prompts the LLM: *"Based on this text, does the author agree with, disagree with, or use the method from the cited paper?"*
5. Graph Construction: The result is saved as a labeled edge in Neo4j.
6. Visualization: The frontend renders the graph, color-coding edges (e.g., Red for Critique, Green for Support).

6. Tools & Technologies

- **Input:** User enters a topic (e.g., "Attention Mechanism").
- **Search & Retrieve:** The Agent fetches the top seed papers via API.
- **Context Extraction:** The system parses the full text to find the "citation window" (the 1-2 sentences surrounding a reference).
- **LLM Reasoning:** The Agent prompts the LLM: *"Based on this text, does the author agree with, disagree with, or use the method from the cited paper?"*
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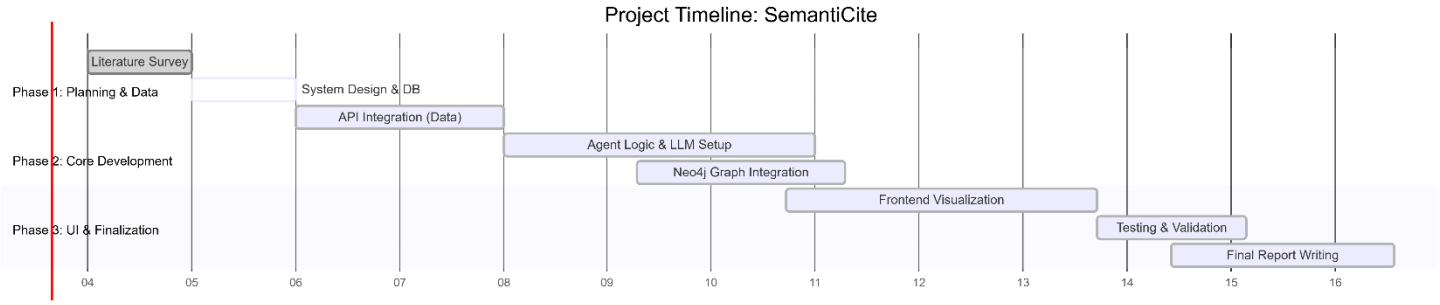
7. Expected Outcome

- **A Functional "Citation Crawler" Agent:** A Python-based autonomous agent capable of retrieving research papers and their citation networks from the Semantic Scholar API without human intervention.
- **A Semantic Knowledge Graph:** A populated Neo4j database containing nodes (Research Papers) and edges (Relationships) labeled with semantic meaning (e.g., *Refutes*, *Supports*, *Extends*).
- **Interactive Web Dashboard:** A user-friendly frontend (built with React/Cytoscape.js) that allows researchers to visualize the "genealogy" of a research topic dynamically.
- **Context-Aware Search Results:** A search mechanism that filters citations not just by relevance, but by "intent" (e.g., "Show me only papers that *critique* the Transformer architecture").

8. Proposed Timeline



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